Monitoring quality drift

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Ask for the denominator, missing count and comparison conditions.

Teach descriptive monitoring and uncertainty with explicit threshold and coverage policies.

Keep one habit: Model main rates → inspect picture slices → compare empty and missing cases → assess new denominator/threshold → record causal limits.

MAIN LESSON · SECTION 1

Prepare the connected service investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of monitoring quality **drift**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Release and rollback. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of release and rollback

Reference example, not the only acceptable wording: Teacher explanation: Release records bind code, configuration, data schema and evaluation **evidence**. Rollback requires compatibility with the current stored data.

Version v2 writes schema 2. Version v1 reads only schema 1. A rollback to v1 after v2 writes data cannot safely proceed without migration or a compatible reader.

Without opening the old lesson, explain one key idea from Queues and asynchronous work. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of queues and asynchronous work

Reference example, not the only acceptable wording: Teacher explanation: Queues separate acceptance from completion. Work may be redelivered, so consumers must preserve operation IDs and deduplicate committed effects.

A queued job is pending, running, completed or failed. Acknowledge only after the declared durable completion step. A repeated completed ID returns its prior result; unknown status needs reconciliation.

Without opening the old lesson, explain one key idea from Sandboxing and resource limits. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of sandboxing and resource limits

Reference example, not the only acceptable wording: Teacher explanation: Sandboxing and quotas constrain what a **process** may access or consume. They reduce possible operations but do not verify every result.

The fictional job may read its fixture directory, write its **output** directory, use no network and run for at most 2 seconds with at most 10 queued tasks. Reject excess work before starting it.

Pictured end goal

Explain monitoring quality **drift** using the deciding step in this fictional service.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Release and rollback

Queues and asynchronous work

Sandboxing and resource limits

Use the three-session teaching cycle above, with extra time as needed. Monitoring compares current outcomes with a declared reference and sampling protocol. A change in observed quality may reflect case mix or instrumentation. Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions. Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of monitoring quality **drift**

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Teach denominators, question slices and evidence limits together

> Distinguish the main 80% rule from the selected independent 70% rule.

Definitions and complete measurement protocol

Monitoring compares current outcomes with a declared reference and sampling protocol. A change in observed quality may reflect case mix or instrumentation.

For each window report eligible counts, passes, missing outcomes and condition mix. Alert when success falls below the toy **80%** threshold; investigate before attributing cause.

Measurement item	Complete fictional classroom protocol
Eligible cases	Supplied count in each window; passes + failures + missing must equal it
Observed success rate	Passes / (passes + observed failures); label this denominator explicitly
Main complete windows	Reference 18 passes / 20; current 14 / 20; no missing outcomes
Main alert threshold	Alert strictly below fictional 80%; exactly 80% does not alert
Missing outcomes in added code probe	Report missing count and coverage alert; suspend full-window quality alert until complete
Empty question slice	No observations: rate unavailable, not 0% or 100%; report coverage gap
Question types	Picture separates clear and ambiguous questions, same supplied rubric
Comparison record	Keep rubric, sampling protocol, source/model versions and condition mix
Interpretation limit	Descriptive supplied counts; no established stable trend, cause or statistical significance

Main rate ledger and checked reasoning

Main window	Passes / eligible	Missing	Success and 80% rule
Reference	18 / 20	0	90%; no below-threshold alert

Main window	Passes / eligible	Missing	Success and 80% rule
Current	14 / 20	0	70%; below-threshold alert
Change	Same denominator; four fewer passes	0	70 − 90 = −20 percentage points
Meaning of alert	Investigate supplied outcome change	No causal evidence added	Check case mix, versions, rubric and instrumentation
Boundary probe	8 / 10	0	80%; equality does not trigger strictly-below alert

Main 18/20 = 90% versus 14/20 = 70%, down 20 percentage points; only the new rate is below the toy 80% threshold. Exactly 80% does not alert. Picture clear 90→95%, ambiguous 80→40%, totals 26/30→23/30, about 86.7→76.7%, down 10 percentage points. With no ambiguous B questions, report unavailable ambiguous rate and changed case mix; total B becomes 19/20, not evidence that ambiguity improved.

Added missing-outcome protocol labels observed-only denominator, raises coverage alert and suspends full-window quality decision. No stable trend or cause follows. Selected independent A 8/10 = 80%, B 12/20 = 60%; B is below its new 70% threshold despite more raw passes.

Monitoring quality drift — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Reference window passes 18/20; new window 14/20.	For each window report eligible counts, passes, missing outcomes and condition mix. Alert when success falls below the toy 80% threshold; investigate before attributing cause.	90% versus 70%, drop 20 percentage points; new window breaches 80%. Check case mix, data versions and logs before asserting model drift as the cause.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: 90% versus **70%**, drop 20 percentage points; new window breaches **80%**. Check case mix, data versions and logs before asserting model **drift** as the cause.

Qualification: An alert identifies a condition for investigation, not a complete causal explanation.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Reference window passes 18/20; new window 14/20.

90% versus **70%**, drop 20 percentage points; new window breaches **80%**. Check case mix, data versions and logs before asserting model **drift** as the cause.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Missingness, sampling and causal probes

Monitoring probe	Expected interpretation
Rate below the declared threshold	Alert and investigate; cause remains unproven
Rate exactly at the declared threshold	No strictly-below alert
More passes with more total questions	Compute rates; raw pass totals alone do not compare quality
Missing outcomes	Report count and observed denominator; do not silently relabel them
Zero eligible observations	Unavailable rate and coverage gap
One slice weakens while another improves	Report both; aggregate alone hides the different directions
Case mix or rubric changes	Comparability needs review
Counts do not sum to eligible total	Reject inconsistent measurement fixture
Small sample change	Does not establish stable trend, significance or model-caused drift

Separate picture counts and full arithmetic

Monitor separate question slices. A service uses the same rubric on clear and ambiguous questions in two recorded weeks.

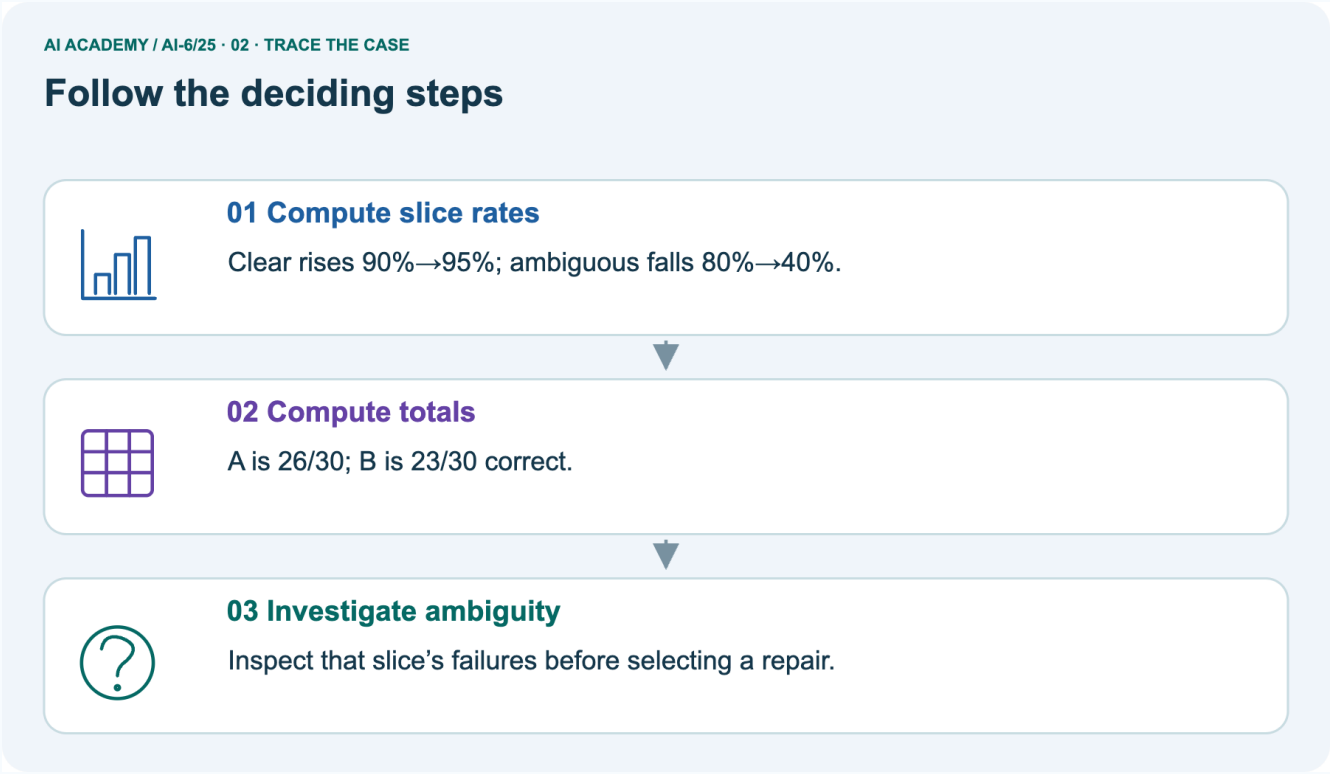
A: clear 18/20, ambiguous 8/10. B: clear 19/20, ambiguous 4/10.

The picture is a separate two-slice case. Clear questions improve from 18/20 = 90% to 19/20 = 95%, up 5 percentage points. Ambiguous questions fall from 8/10 = 80% to 4/10 = 40%, down 40 percentage points. Totals are 26/30 ≈ 86.7% and 23/30 ≈ 76.7%, down 10 percentage points, with these complete counts and fixed slice sizes. The main example's 18/20 and 14/20 are different windows.

If pictured week B instead has no ambiguous questions, that slice has 0 eligible observations and an unavailable rate, not 0% success. Its observed total becomes 19/20 = 95%, but changed case mix prevents interpreting that aggregate rise as improved ambiguous-question performance.

Picture slice	Recorded week A	Recorded week B	Change in percentage points
Clear	18/20 = 90%	19/20 = 95%	+5
Ambiguous	8/10 = 80%	4/10 = 40%	−40
All supplied cases	26/30 ≈ 86.7%	23/30 ≈ 76.7%	−10
Changed B: no ambiguous observations	A stays unchanged	Clear 19/20; ambiguous 0/0 unavailable	Aggregate 95% now has different case mix

Picture trace and empty-slice qualification



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Compute slice rates	Clear rises 90%→95%; ambiguous falls 80%→40%.
2. Compute totals	A is 26/30; B is 23/30 correct.
3. Investigate ambiguity	Inspect that slice's failures before selecting a repair.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Slice monitoring exposes changes hidden by common easy questions. Retain each denominator and compare the measurement conditions.

Small samples do not establish a stable trend or cause; retain uncertainty and compare conditions.

Change: Week B has no ambiguous questions.

Prediction: That slice is unmeasured, not perfectly correct.

Reason: No **observation**s means no rate can be calculated for it.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Monitoring window	The declared interval or batch for measurement
Drift	A change in data or behaviour relative to a reference
Alert threshold	A condition triggering investigation

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Monitoring quality drift

The facts we will use

AI ACADEMY / AI-6/25

Quality slices: retain denominators

	Clear correct	Clear total	Ambiguous correct	Ambiguous total
Week A	18	20	8	10
Week B	19	20	4	10

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-6/25

Monitoring quality drift

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute slice rates

Clear rises 90% -> 95%; ambiguous falls 80% -> 40%.



2. Compute totals

A is 26/30; B is 23/30 correct.



3. Investigate ambiguity

Inspect that slice's failures before selecting a repair.

CHANGE ONE THING

Week B has no ambiguous questions.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Reference window passes 18/20; new window 14/20. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: 90% versus **70%**, drop 20 percentage points; new window breaches **80%**. Check case mix, data versions and logs before asserting model **drift** as the cause.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Monitoring quality **drift**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: An alert identifies a condition for investigation, not a complete causal explanation.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Compute both clear and ambiguous slice rates for A and B. Which slice weakens, and how should that slice be reported when no ambiguous questions are observed?

Clear rises 90%→95%; ambiguous falls **80%**→40%. A is 26/30; B is 23/30 correct. Inspect that slice's failures before selecting a repair. Changed case: That slice is unmeasured, not perfectly correct.

No **observation**s means no rate can be calculated for it.

Changed-case reasoning: That slice is unmeasured, not perfectly correct. No **observation**s means no rate can be calculated for it.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main 90% versus 70% drops 20 percentage points and breaches the toy 80% rule. Picture clear improves 5 points while ambiguous falls 40 points; totals fall 10 points. Empty ambiguous slice is unavailable, not zero success. Missing-outcome coverage and full-window quality decision are separate under the added declared protocol.

Selected independent case: 8/10 = 80%, 12/20 = 60%; B breaches its supplied 70% threshold even though its raw pass count rose. Investigate case mix, versions and instrumentation before asserting cause.

Diagnose the current misconceptions

The worked result establishes every future case: An alert identifies a condition for investigation, not a complete causal explanation.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows	Next teaching action
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new service scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Window A passes 8/10; B passes 12/20. Threshold **70%**. Compare.

Teacher explanation: **80%** versus **60%**; B breaches **70%**. Raw pass total rose but success rate fell.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: Monitoring compares current outcomes with a declared reference and sampling protocol. A change in observed quality may reflect case mix or instrumentation.

For each window report eligible counts, passes, missing outcomes and condition mix. Alert when success falls below the toy **80%** threshold; investigate before attributing cause.

W2 Trace the worked case

Reference window passes 18/20; new window 14/20. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: 90% versus **70%**, drop 20 percentage points; new window breaches **80%**. Check case mix, data versions and logs before asserting model **drift** as the cause.

W3 Audit the limit

A report omits this qualification: An alert identifies a condition for investigation, not a complete causal explanation. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: An alert identifies a condition for investigation, not a complete causal explanation.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. An alert identifies a condition for investigation, not a complete causal explanation.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: 90% versus **70%**, drop 20 percentage points; new window breaches **80%**. Check case mix, data versions and logs before asserting model **drift** as the cause.

Qualification: An alert identifies a condition for investigation, not a complete causal explanation.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Window A passes 8/10; B passes 12/20. Threshold **70%**. Compare.

Teacher explanation: **80%** versus **60%**; B breaches **70%**. Raw pass total rose but success rate fell.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: **80%** versus **60%**; B breaches **70%**. Raw pass total rose but success rate fell. Require intermediate reasoning and the stated scope. Fresh retry: Window 10/12 then 12/12 under 90% threshold.

Expected reasoning: 83.333% breaches; 100% meets. These small windows do not establish a stable long-term rate.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit service design

> Use saved evidence and boundary decisions to choose the next step.

Project and next-week bridge

Add a labelled monitoring quality **drift evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of monitoring quality **drift**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Incident investigation. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Delayed return and support record

Suggested return points in this level: ai-6/26, ai-6/28, ai-6/31. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	

Record	Teacher evidence
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of monitoring quality **drift**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: Reference window passes 18/20; new window 14/20. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: 90% versus **70%**, drop 20 percentage points; new window breaches **80%**. Check case mix, data versions and logs before asserting model **drift** as the cause.

Check every learner before discussion: Use the worked case for Monitoring quality **drift**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: An alert identifies a condition for investigation, not a complete causal explanation.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Week B has no ambiguous questions. Require a **prediction** and a reason before comparing. Expected result: That slice is unmeasured, not perfectly correct.

Reason: No **observation**s means no rate can be calculated for it.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

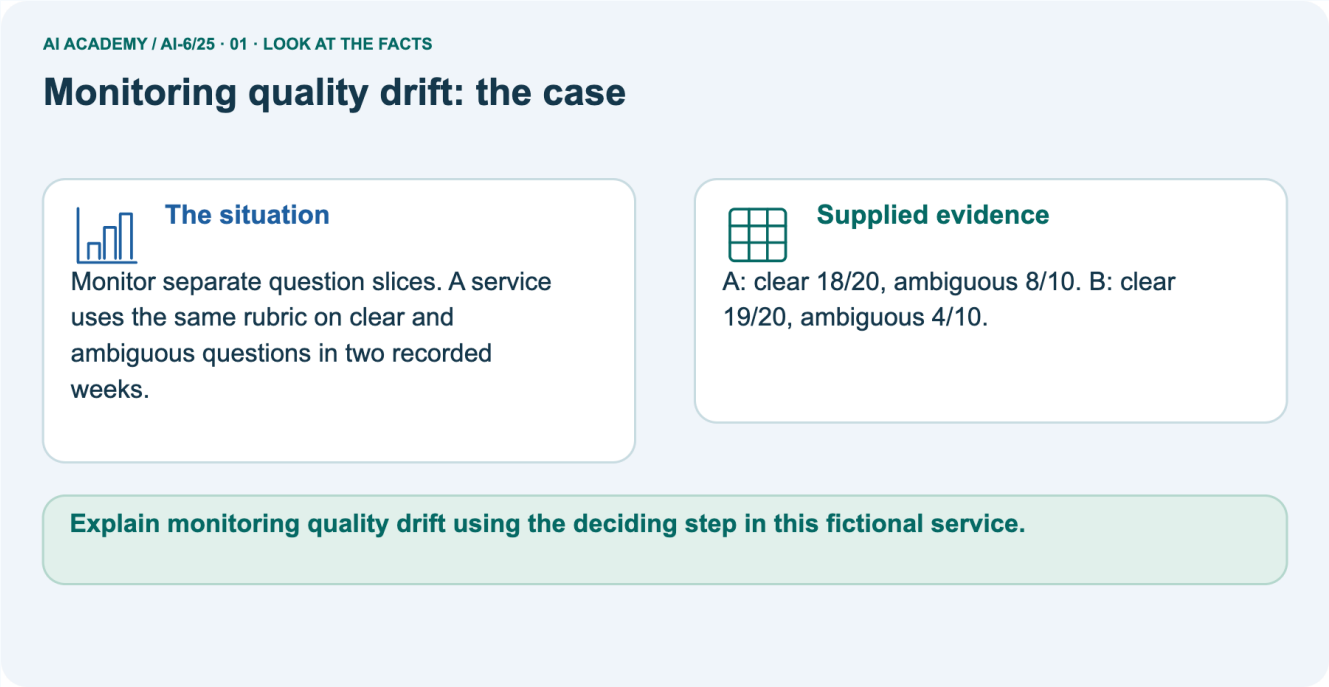
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Monitoring quality drift: the case



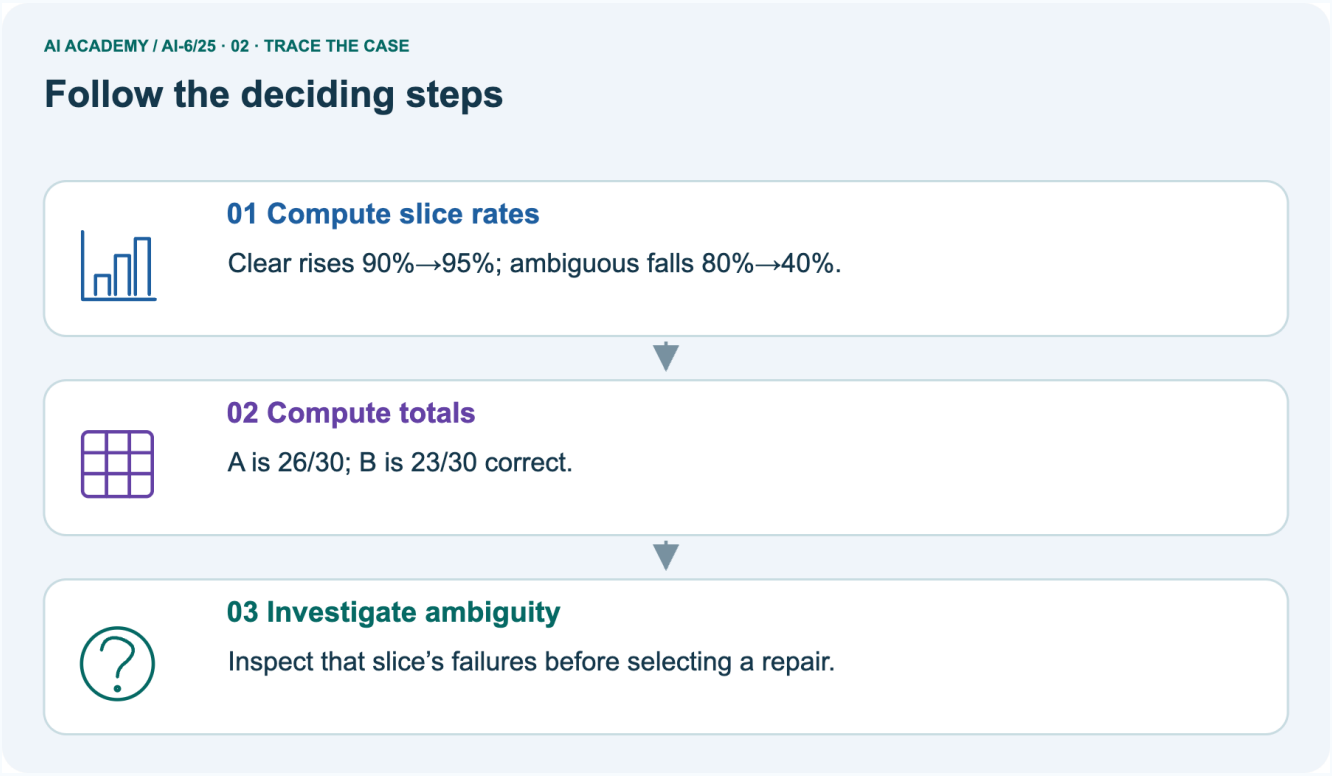
Follow the labelled picture with the matching explanation. Open the vector diagram

Monitor separate question slices. A service uses the same rubric on clear and ambiguous questions in two recorded weeks. Supplied facts: A: clear 18/20, ambiguous 8/10. B: clear 19/20, ambiguous 4/10.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



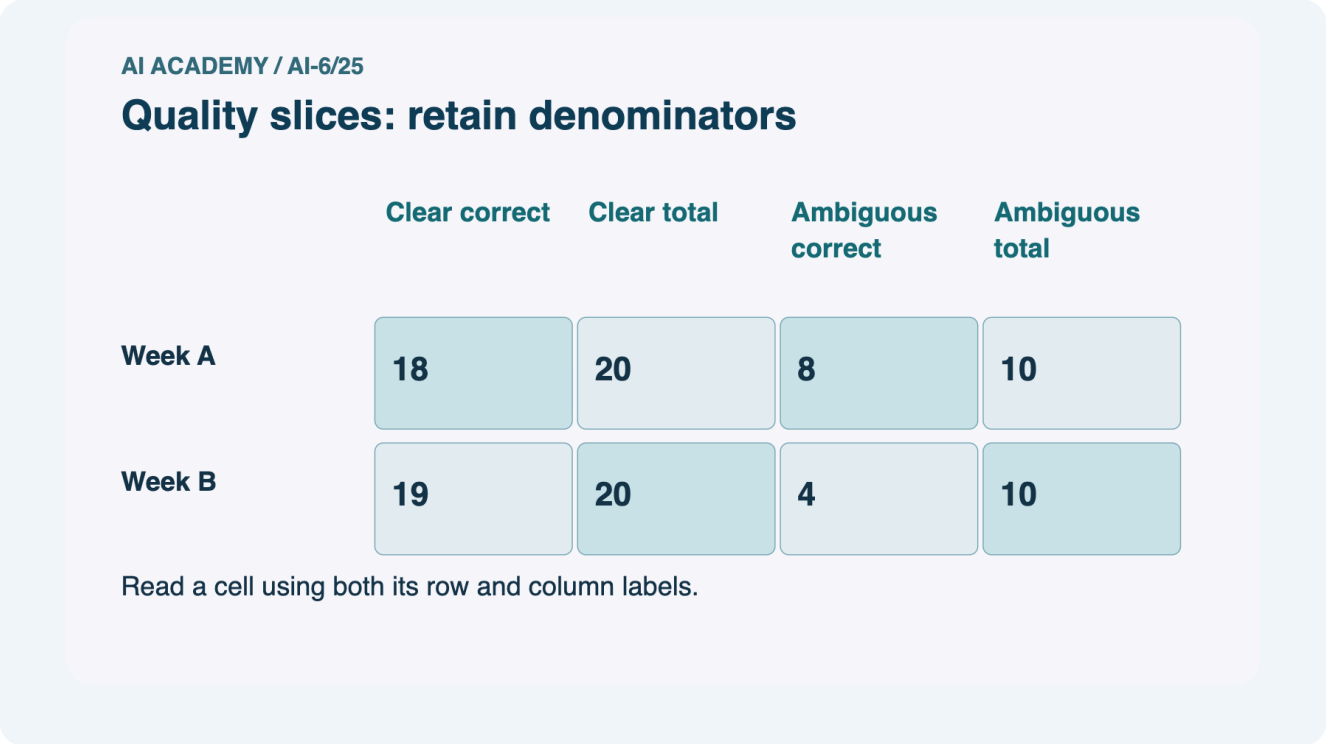
Follow the labelled picture with the matching explanation. Open the vector diagram

Compute slice rates: Clear rises 90%→95%; ambiguous falls **80%**→40%. → Compute totals: A is 26/30; B is 23/30 correct. → Investigate ambiguity: Inspect that slice's failures before selecting a repair.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Quality slices: retain denominators



Follow the labelled picture with the matching explanation. Open the vector diagram

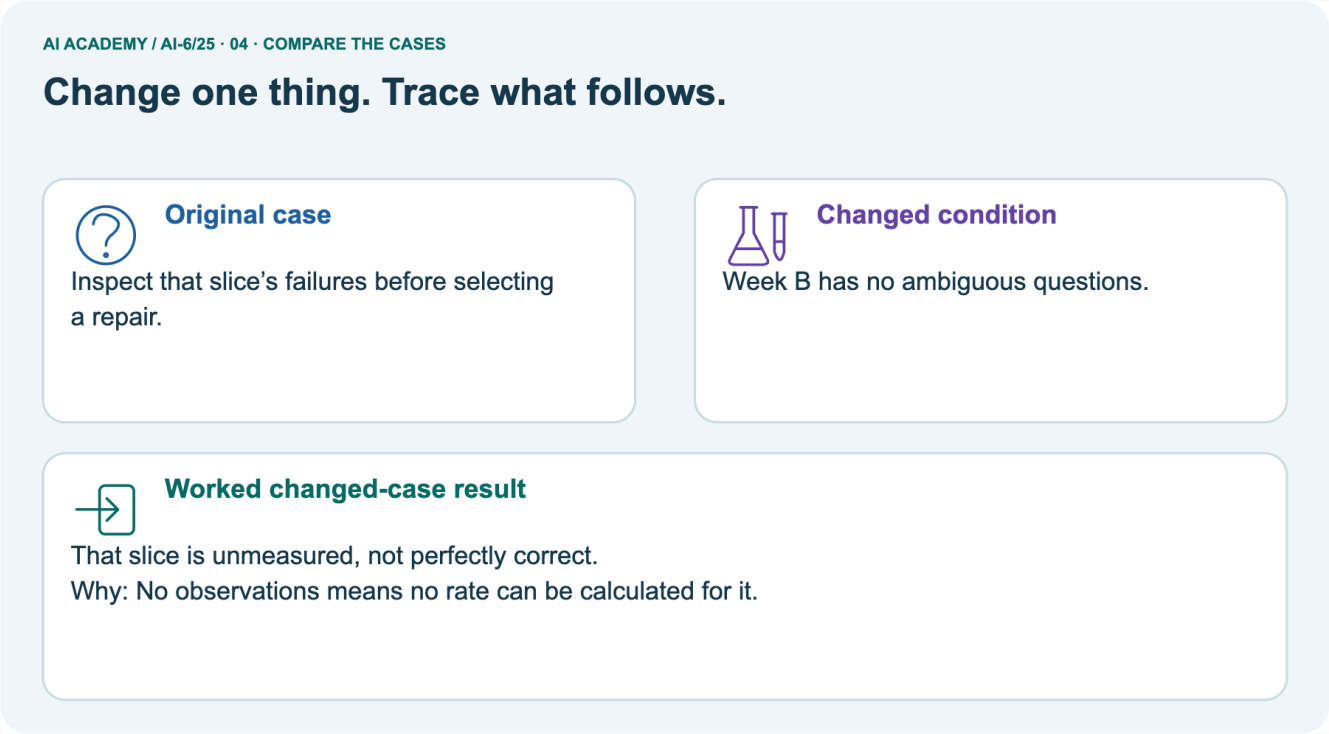
Quality slices: retain denominators. Week A: Clear correct = 18, Clear total = 20, Ambiguous correct = 8, Ambiguous total = 10. Week B: Clear correct = 19, Clear total = 20, Ambiguous correct = 4, Ambiguous total = 10.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Inspect that slice's failures before selecting a repair. Changed condition: Week B has no ambiguous questions. Expected result: That slice is unmeasured, not perfectly correct.

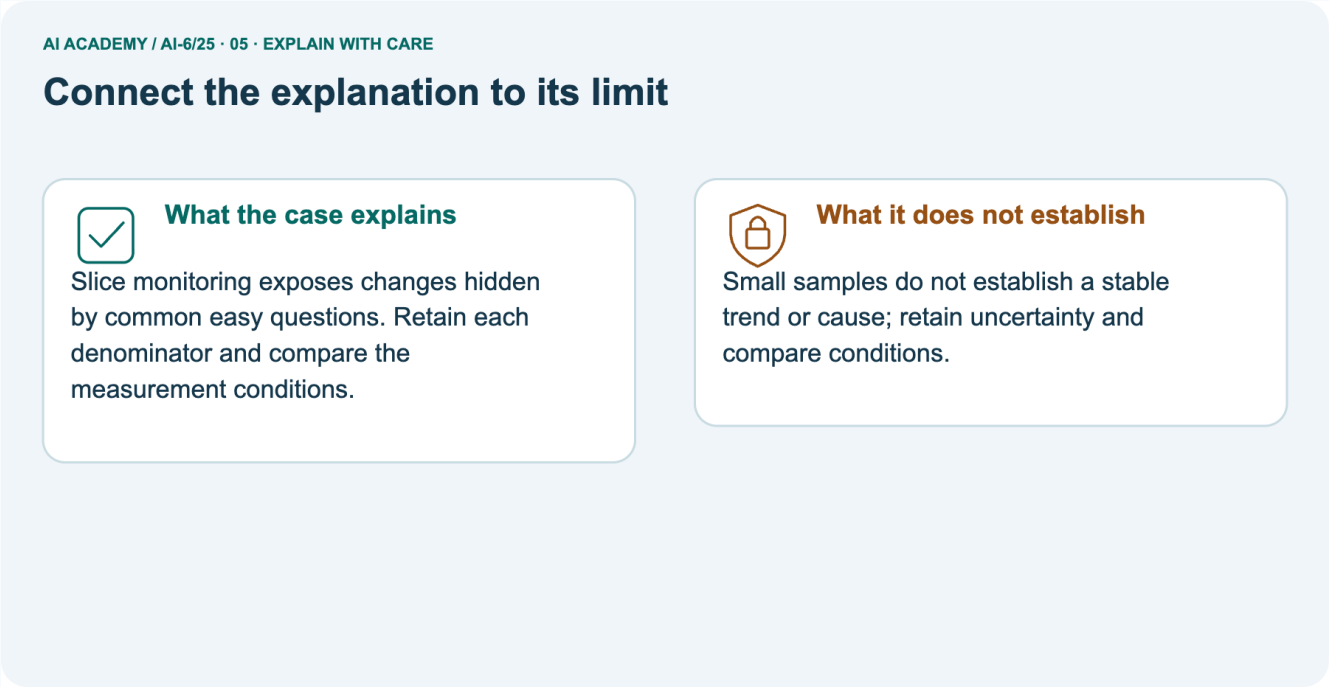
Reason: No **observation**s means no rate can be calculated for it.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: No **observation**s means no rate can be calculated for it.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Slice monitoring exposes changes hidden by common easy questions. Retain each denominator and compare the measurement conditions.

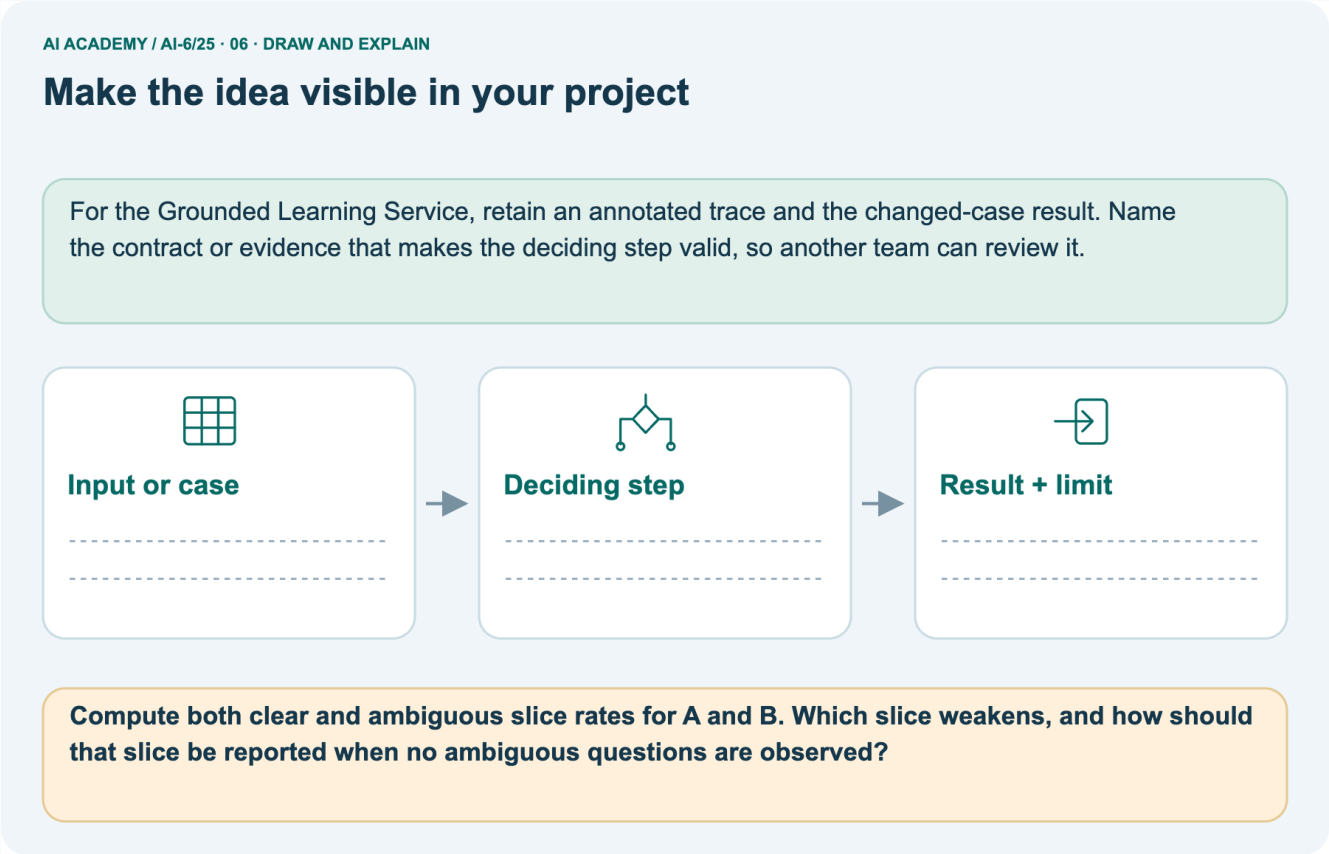
Limit: Small samples do not establish a stable trend or cause; retain uncertainty and compare conditions.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Small samples do not establish a stable trend or cause; retain uncertainty and compare conditions.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Compute both clear and ambiguous slice rates for A and B.

Which slice weakens, and how should that slice be reported when no ambiguous questions are observed?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Clear rises 90%→95%; ambiguous falls **80%**→40%. A is 26/30; B is 23/30 correct. Inspect that slice's failures before selecting a repair. Changed case: That slice is unmeasured, not perfectly correct.

No **observation**s means no rate can be calculated for it.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Monitoring quality drift: the case

02. Follow the deciding steps

03. Quality slices: retain denominators

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.